

**Answer ALL questions.**

**Write your answers in the spaces provided.**

- 1** Vitamin C is a dietary antioxidant.

Spinach is a plant that produces edible leaves with a high vitamin C content.

- (a) Explain the link between dietary antioxidants and the risk of cardiovascular disease (CVD).

(3)

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- (c) The effect of soil type and season on the vitamin C content of spinach plants was investigated.

The table shows some data from spinach grown in the two soil types, in spring and autumn.

|                    | Vitamin C content of leaves / $\text{mg g}^{-1}$ |             |             |             |
|--------------------|--------------------------------------------------|-------------|-------------|-------------|
|                    | Spring                                           |             | Autumn      |             |
|                    | Soil type 1                                      | Soil type 2 | Soil type 1 | Soil type 2 |
| Mean               | 0.49                                             | 0.73        | 0.59        | 0.95        |
| Standard deviation | $\pm 0.07$                                       | $\pm 0.09$  | $\pm 0.06$  |             |

- (i) The table shows the data required to calculate the missing standard deviation for the spinach grown in soil type 2 in the autumn.

| $(x - \bar{x})^2$ |
|-------------------|
| 0.031             |
| 0.009             |
| 0.031             |
| 0.004             |
| 0.013             |
| 0.006             |
| 0.001             |
| 0.024             |
| 0.000             |
| 0.024             |
| 0.055             |
| 0.002             |



Calculate the standard deviation ( $s$ ) using this data and the formula:

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

Give your answer to an appropriate number of decimal places.

(3)

Answer .....

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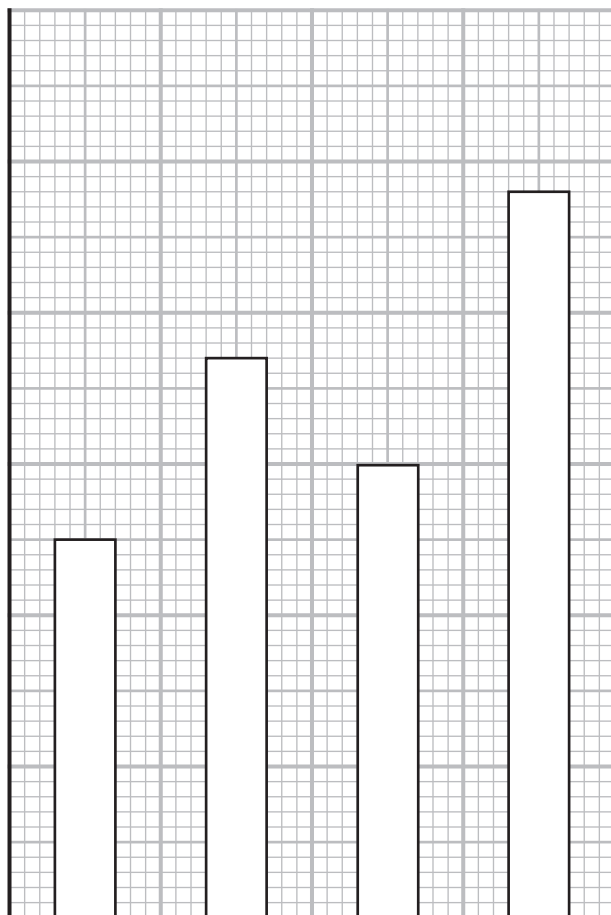
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(ii) The graph shows the means from this investigation.

Complete this graph by adding the axes, values and labels, together with the standard deviations.

(3)



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- (iii) It was concluded that the vitamin C content of the leaves is affected by the soil type and season in which the plants are grown.

Comment on this conclusion.

(3)

**(Total for Question 1 = 17 marks)**

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